

Designing and Implementing Enterprise-Scale Analytics Solutions

Using Microsoft Azure and Microsoft Power BI



Course introduction



Azure Enterprise Data Analyst Role



Subject matter expertise in **designing**, **creating**, and **deploying** enterprise-scale data analytics solutions.

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Expertise **building and implementing Power BI assets**, including managing data repositories and data processing in the cloud and on-premises, along with using Power Query and Data Analysis Expressions (DAX).

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Experience managing a data analytics environment, querying and transforming data, implementing and managing data models, and exploring and visualizing data using **Microsoft Purview, Azure Synapse Analytics, and Power BI.**

Prerequisites



Successful Azure enterprise data analysts start this role with a **foundational knowledge of core data concepts** and how they're implemented using Azure data services.

Prerequisites

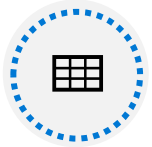


Successful Azure enterprise data analysts start this role with a **foundational knowledge of core data concepts** and how they're implemented using Azure data services.



Experience **designing and building scalable data models, cleaning and transforming data, and enabling advanced analytic capabilities** that provide meaningful business value using Microsoft Power BI.

Course Objectives



Query and transform data.



Implement and manage data models.



Explore and visualize data.



Implement and manage a data analytics environment.

Course outline

Sections

1 - Introduction to data analytics on Azure

2 - Govern data across an enterprise

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- 1 - Introduction to data analytics on Azure
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 - 3 - Model, query, and explore data in Azure Synapse
 - 4 - Prepare data for tabular models in Power BI
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5 - Design and build tabular models

6 - Implement advanced data visualization techniques using Power BI

Course outline

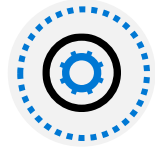
Sections

- 1 - Introduction to data analytics on Azure
 - 2 - Govern data across an enterprise
 - 3 - Model, query, and explore data in Azure Synapse
 - 4 - Prepare data for tabular models in Power BI
 - 5 - Design and build tabular models
 - 6 - Implement advanced data visualization techniques using Power BI
 - 7 - Implement and manage an analytics environment
 - 8 - Manage the analytics development lifecycle
-

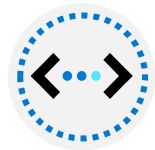
Agenda



Explore Azure data services for modern analytics



Understand concepts of data analytics



Explore data analytics at scale

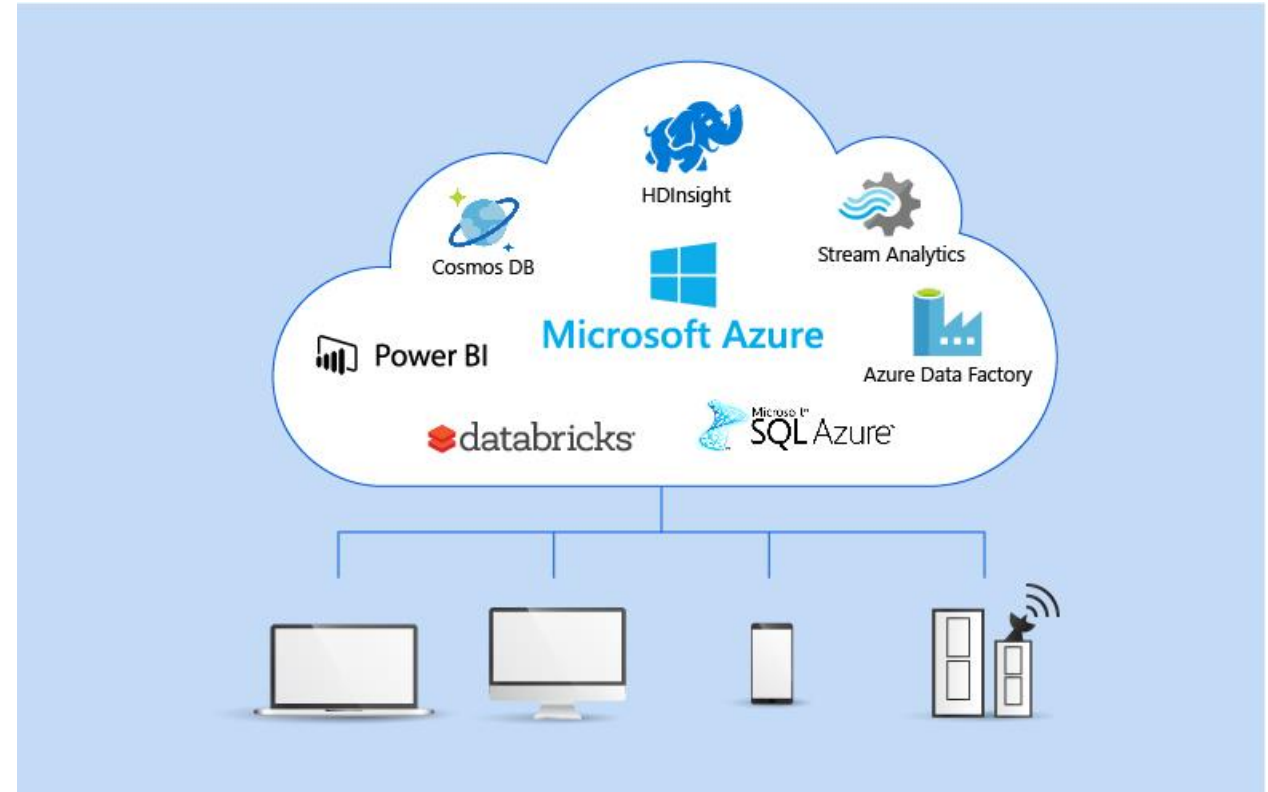
Explore Azure data services for modern analytics



Introduction

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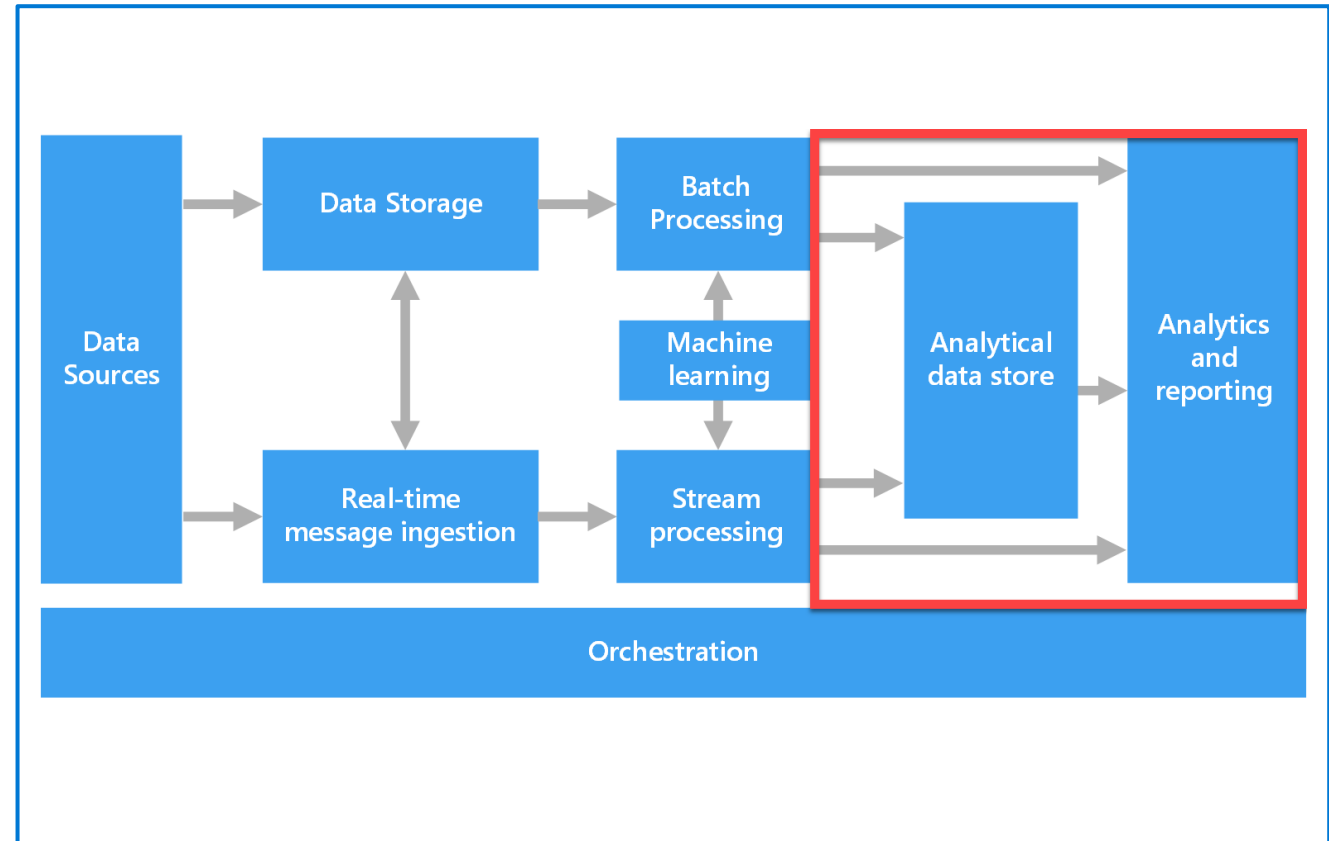
- Explore the Azure data ecosystem for analytics



Understand the Azure data ecosystem

Azure data storage solutions:

- Azure **Storage accounts** are the base storage type within Azure.

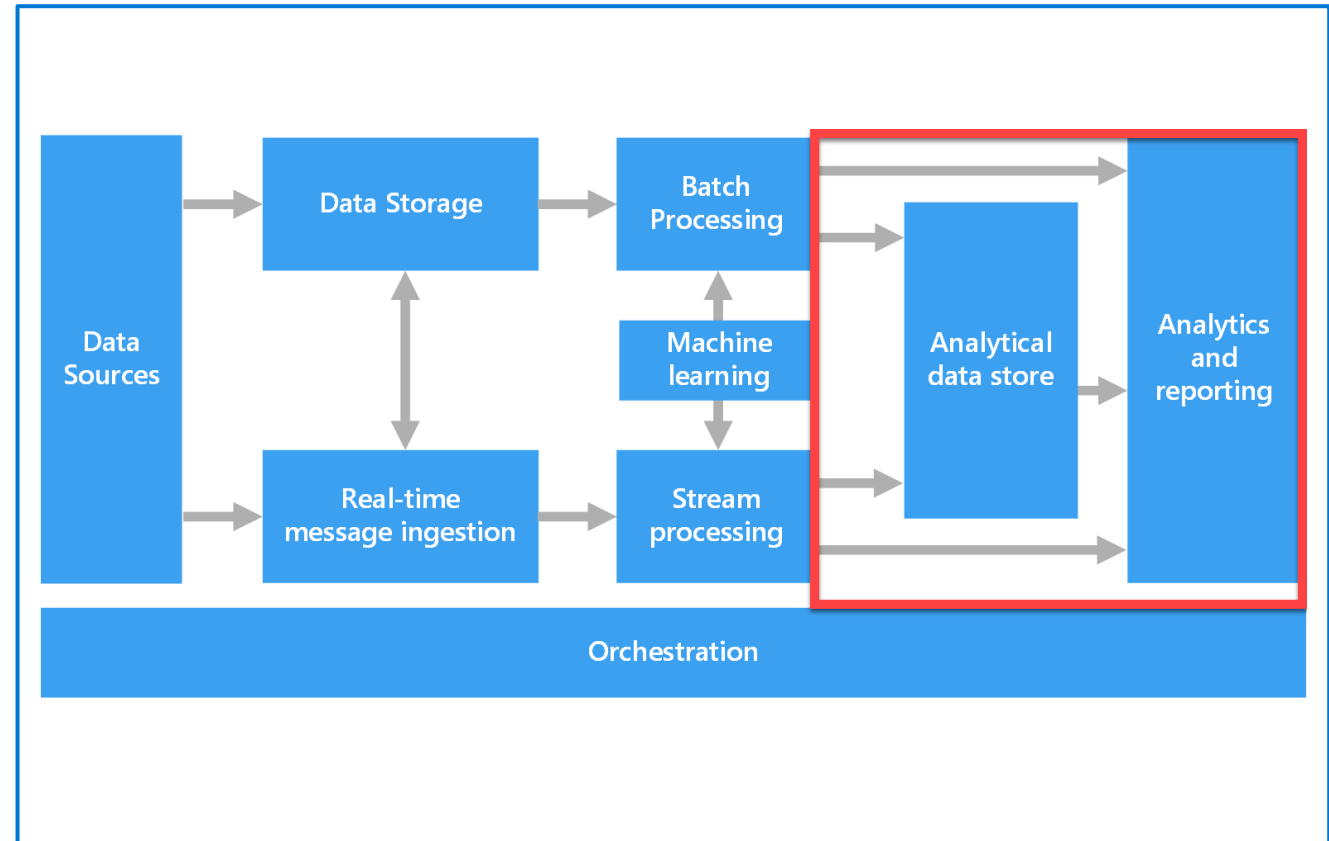


The figure represents the overall data pipeline diagram beginning with data sources on the left and flowing through to analytics and reporting.

Understand the Azure data ecosystem

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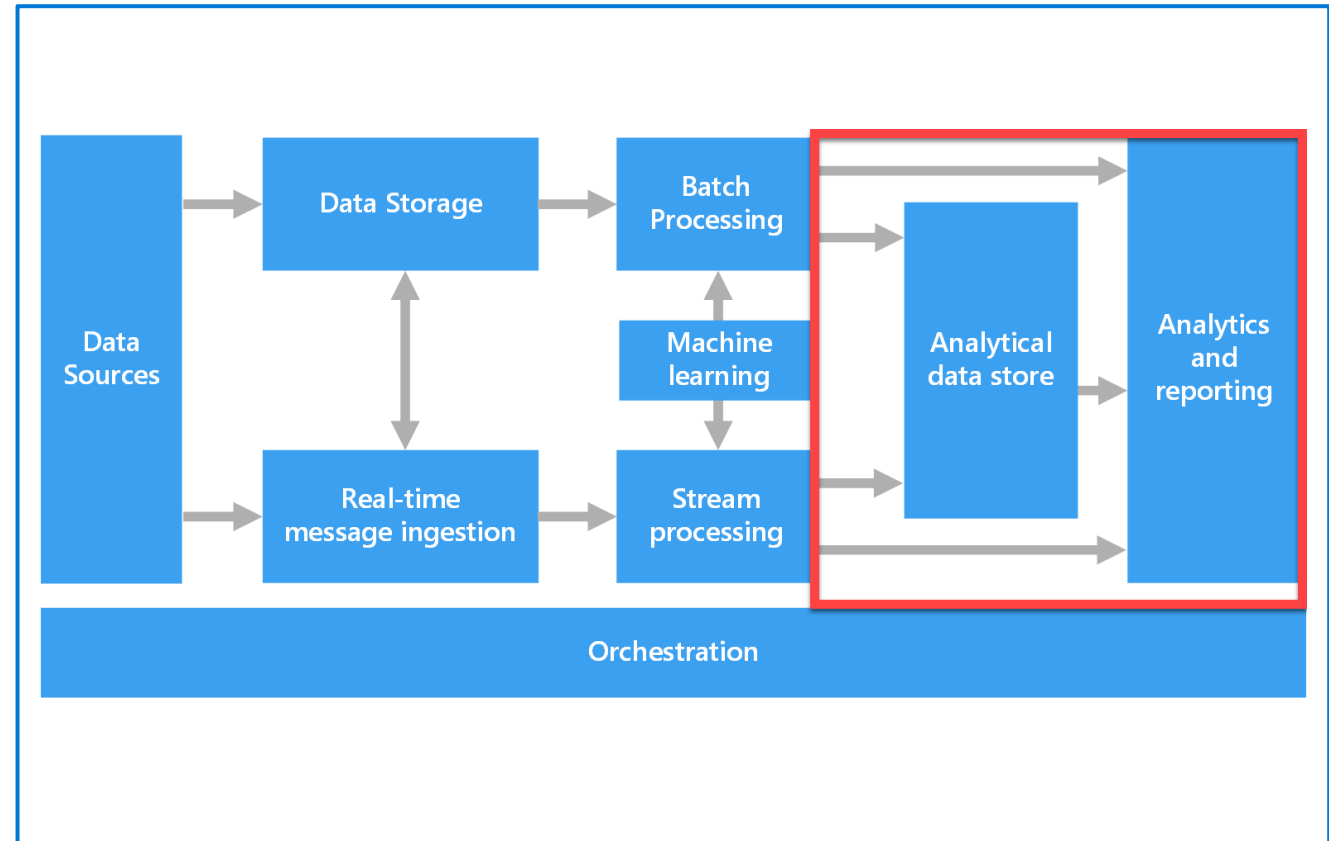
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- Azure Storage offers a **scalable object store** for data objects and file system services in the cloud.



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Understand the Azure data ecosystem

Areas outlined in **red highlight the pieces** of the **analytics solution** that data analysts use to make sense of the data.

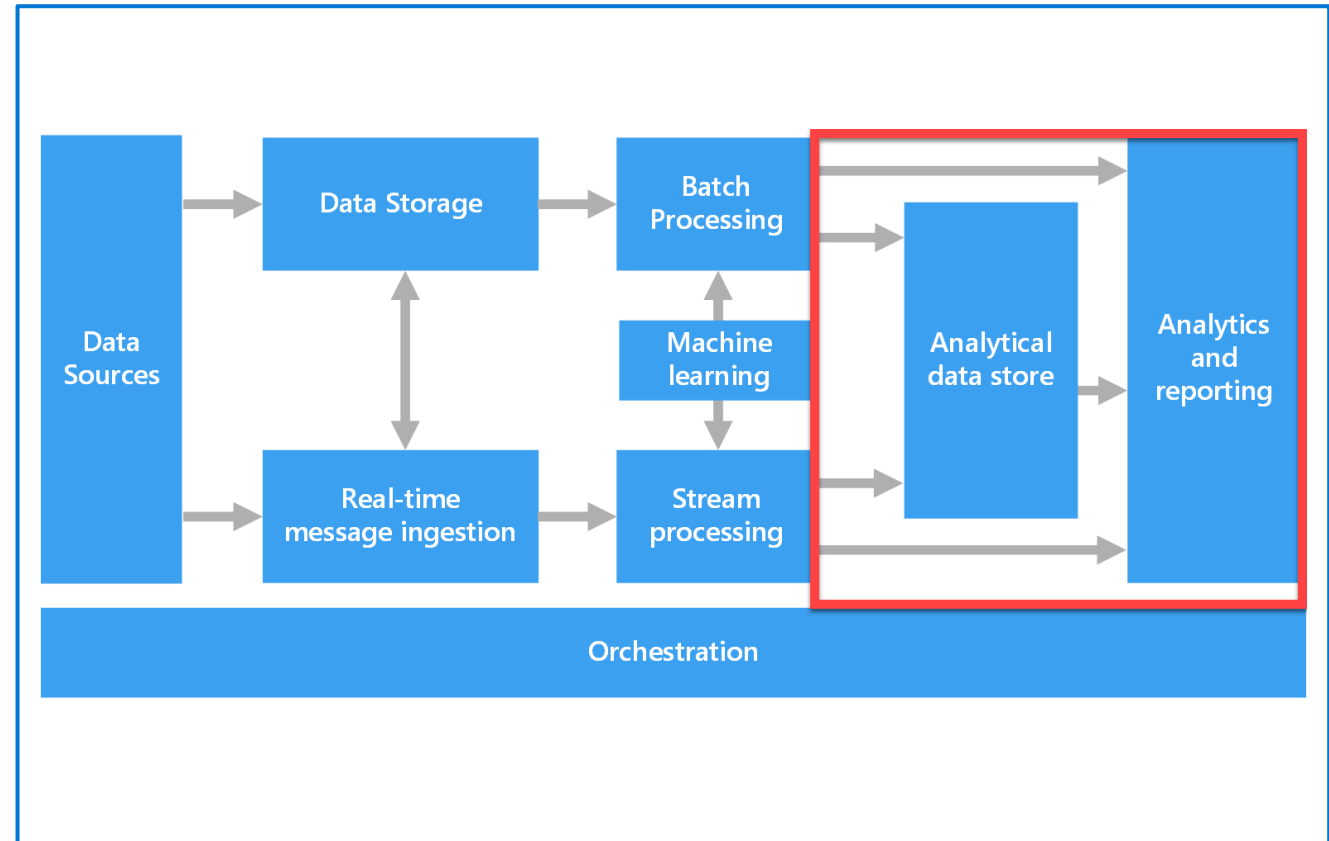


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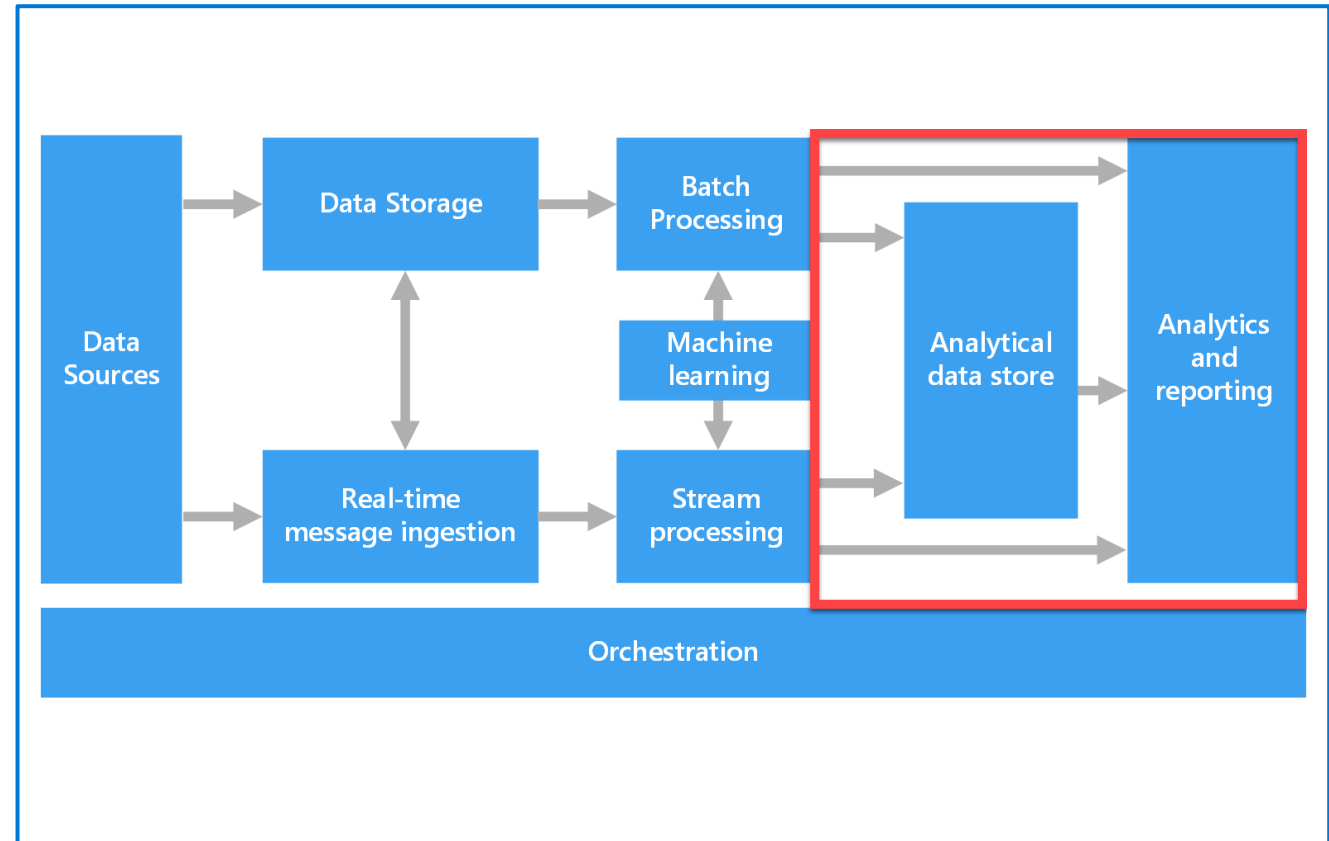
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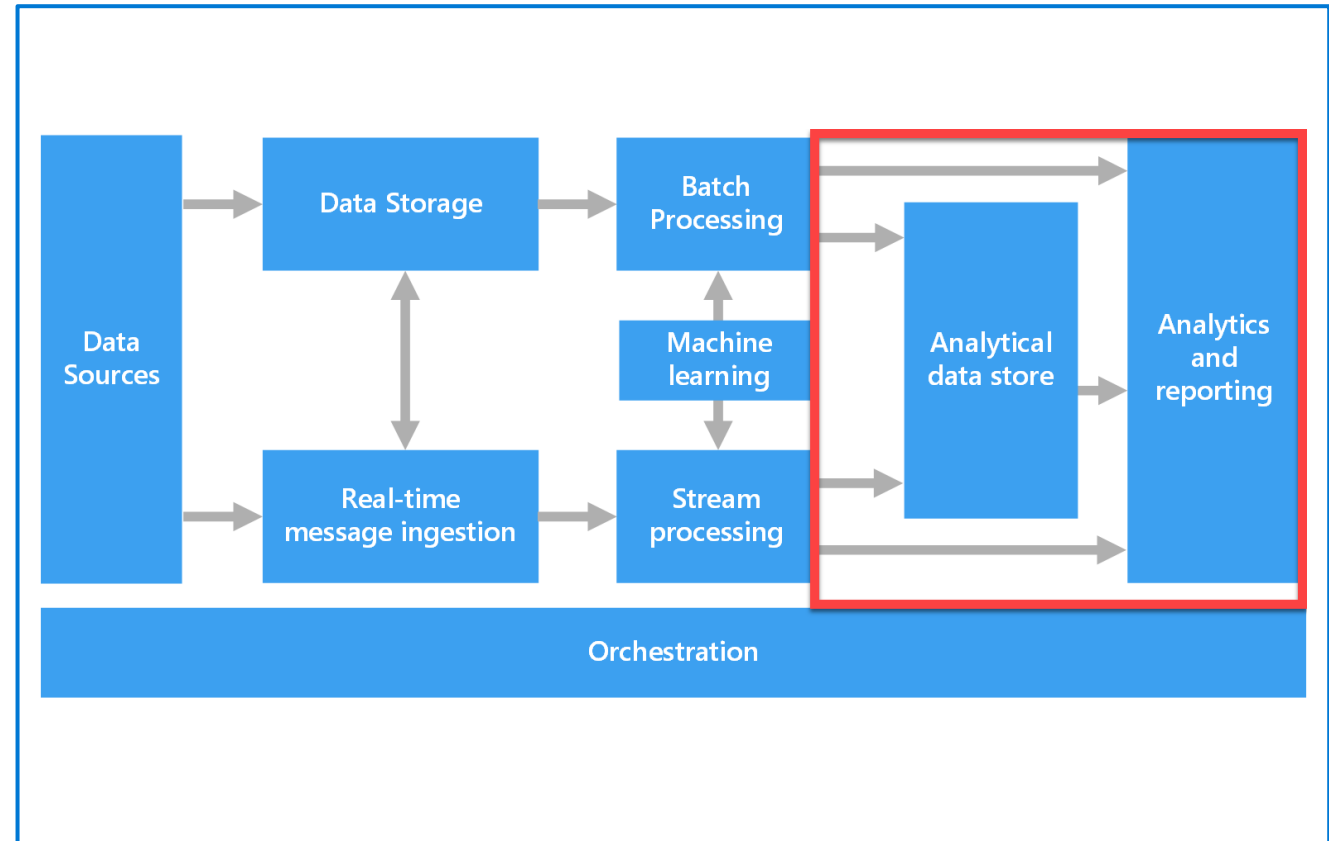
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Areas outlined in **red highlight the pieces** of the **analytics solution** that data analysts use to make sense of the data.

- **Batch** processing is the most common type of data processing, best suited for:
 - large datasets or
 - data coming from legacy data systems.
- Batch processing is **not suited** for **rapid analysis and decision making**.
- **Stream processing** is ideal for projects that require **real-time analysis** and is less suited for projects requiring complex analytics.



While data processing typically occurs **upstream of the analytical data store**, it's critical that analysts understand how data are ingested and at what frequency, to build the appropriate analytics solution.

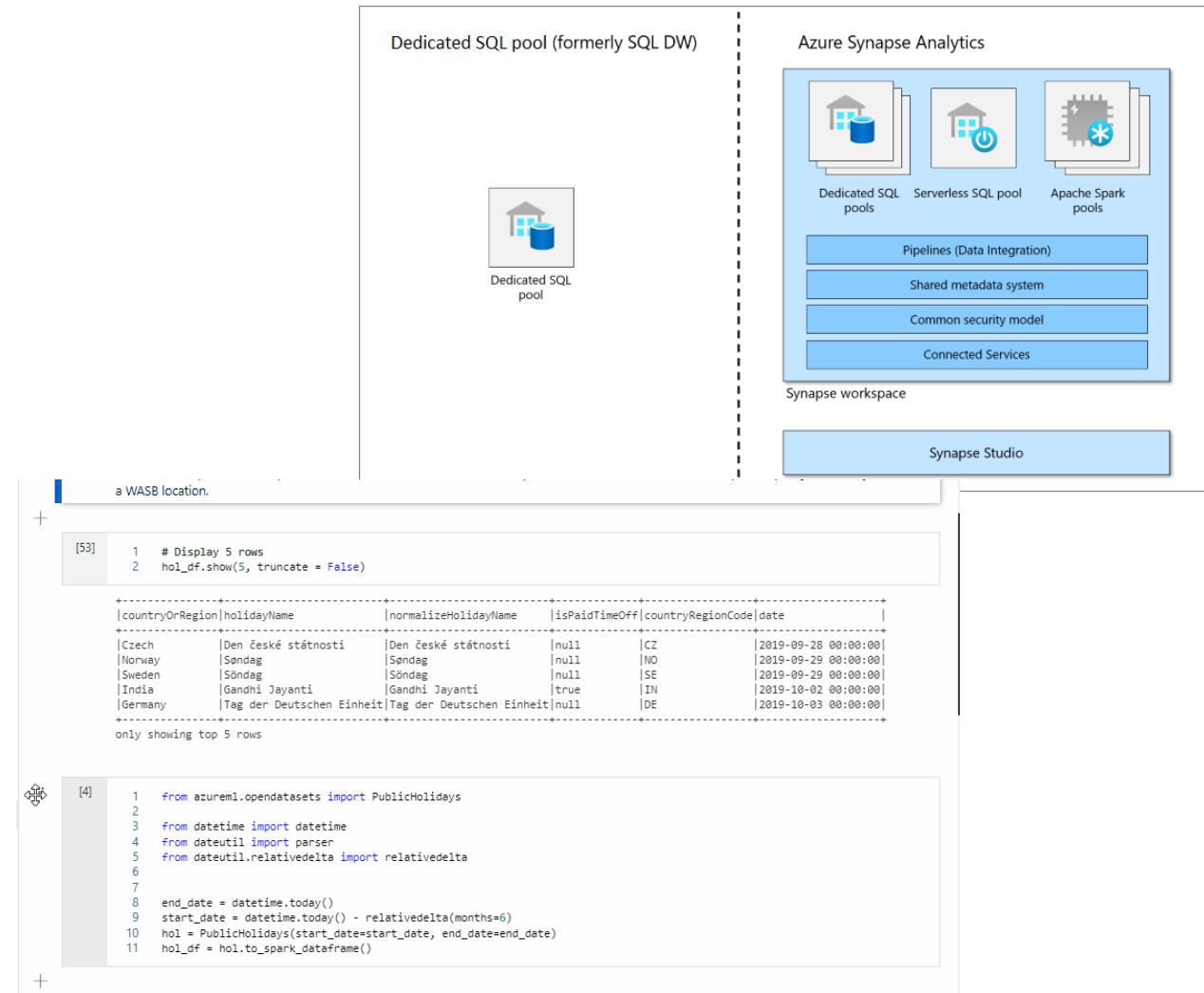
Explore modern analytics solution architecture

Azure data analytics and reporting:

- There are **many options for analytics** and reporting in Azure.

Several tools for exploring and visualizing data in Azure Synapse Analytics :

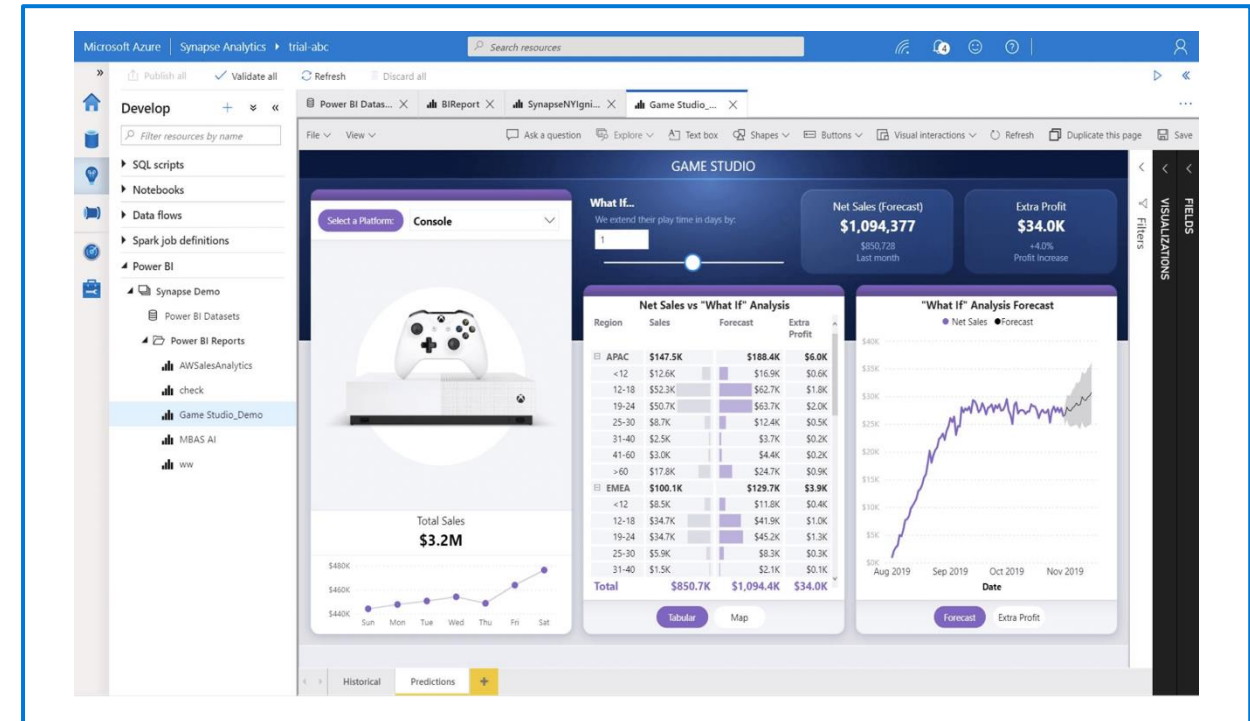
- The Azure Synapse SQL results pane
- Synapse Studio
- Synapse notebooks



Explore modern analytics solution architecture

Visualize data and create reporting solutions using **Microsoft Power BI**:

- **Power BI** is an enterprise analytics tool

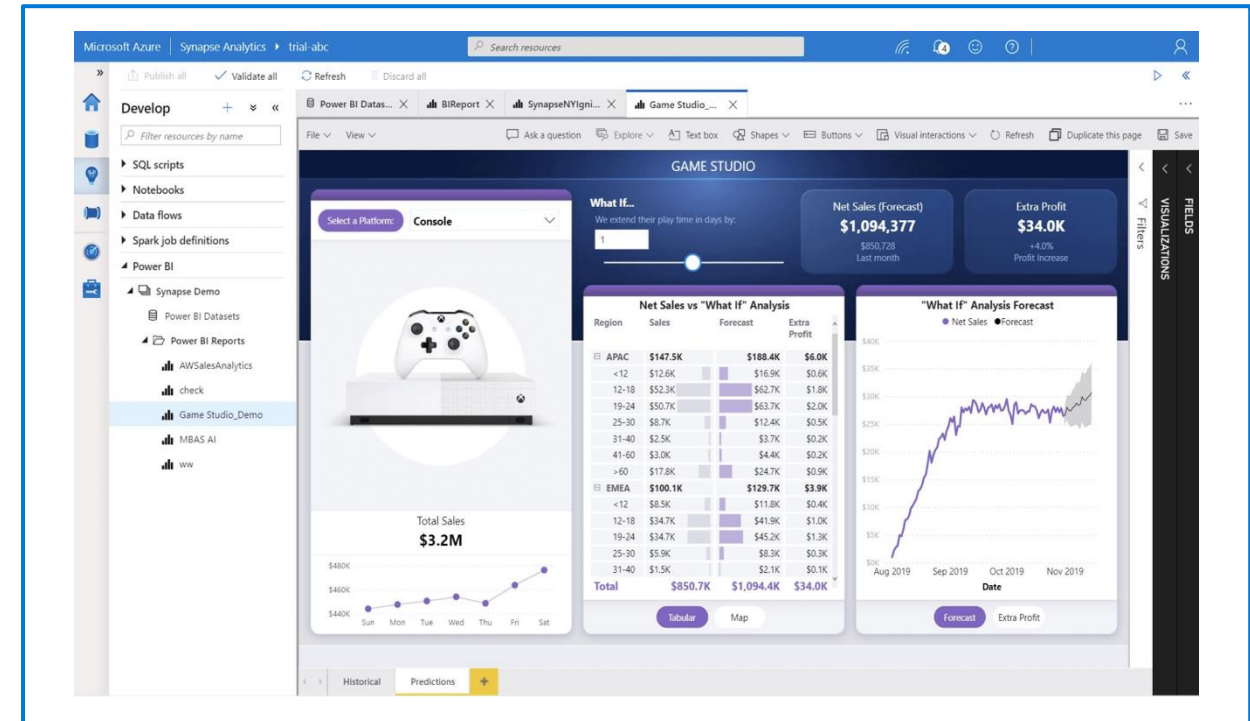


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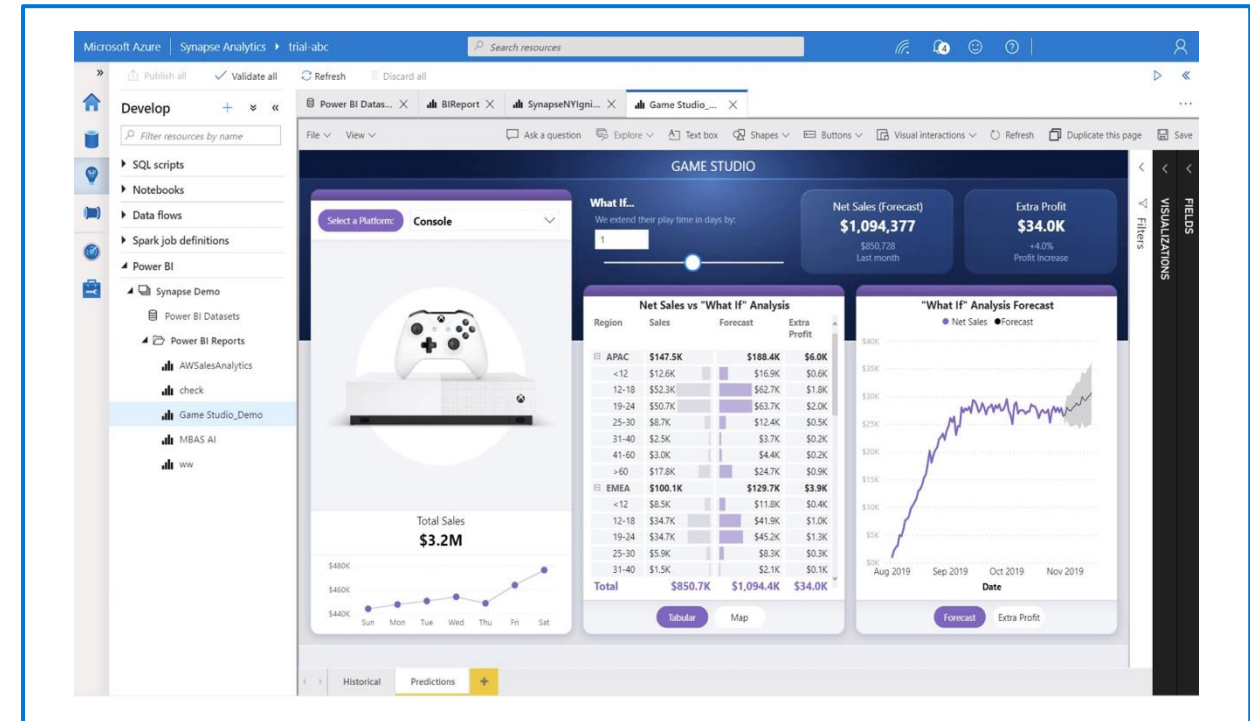


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- Power BI also has **native connectors to many Azure data services**



The figure shows the Power BI Report interface in **Azure Synapse Analytics Workspace**.

Knowledge check



What type of data is best suited for batch processing?

- IoT gas sensor data that monitors and detects the presence of toxic or hazardous gasses.
- Real-time personalized advertising data.
- ➔ Daily sales report data.



What is one advantage of using Power BI over a Synapse notebook for data visualization?

- There is no advantage, Power BI and Azure Synapse Analytics are comparable for data visualization.
- ➔ Power BI was designed for distribution of data products outside of the data team.
- Power BI displays simple visualizations compared to Synapse notebooks.

Summary

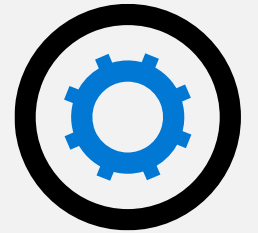
In this section you've learned how to:

- Describe the Azure data ecosystem for analytics

Want to learn more!

- [Explore fundamentals of modern data warehousing](#)
- [Azure data architecture guide](#)

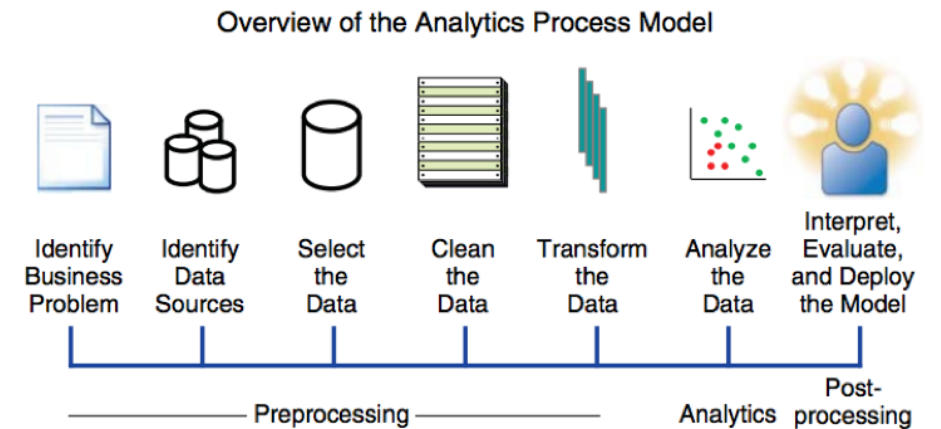
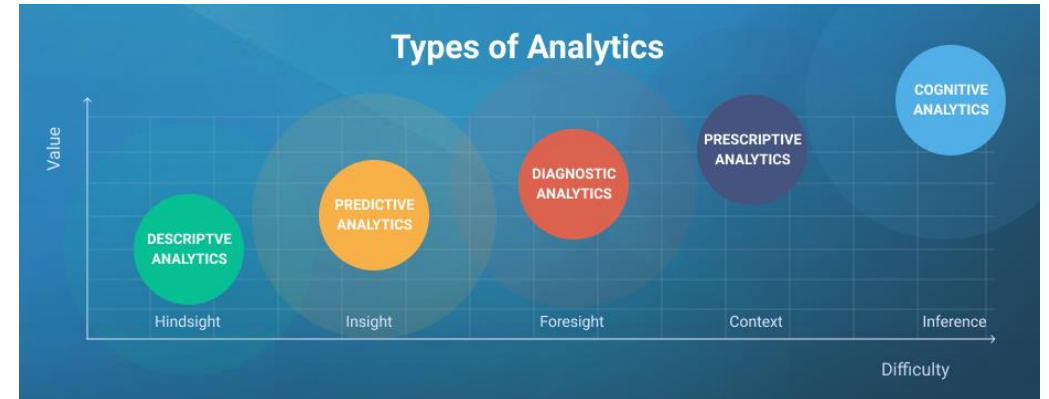
Understand concepts of data analytics



Introduction

In this section you will:

- Define the five **types of data analytics**
- Describe the **data analytics process**
- **Identify data** types and **storage**



Understand data analytics types

Data analytics is used for exploring data, extracting insights, and acting on those insights.

Data analytics covers a **range of activities**, each with its **own focus and goals**.

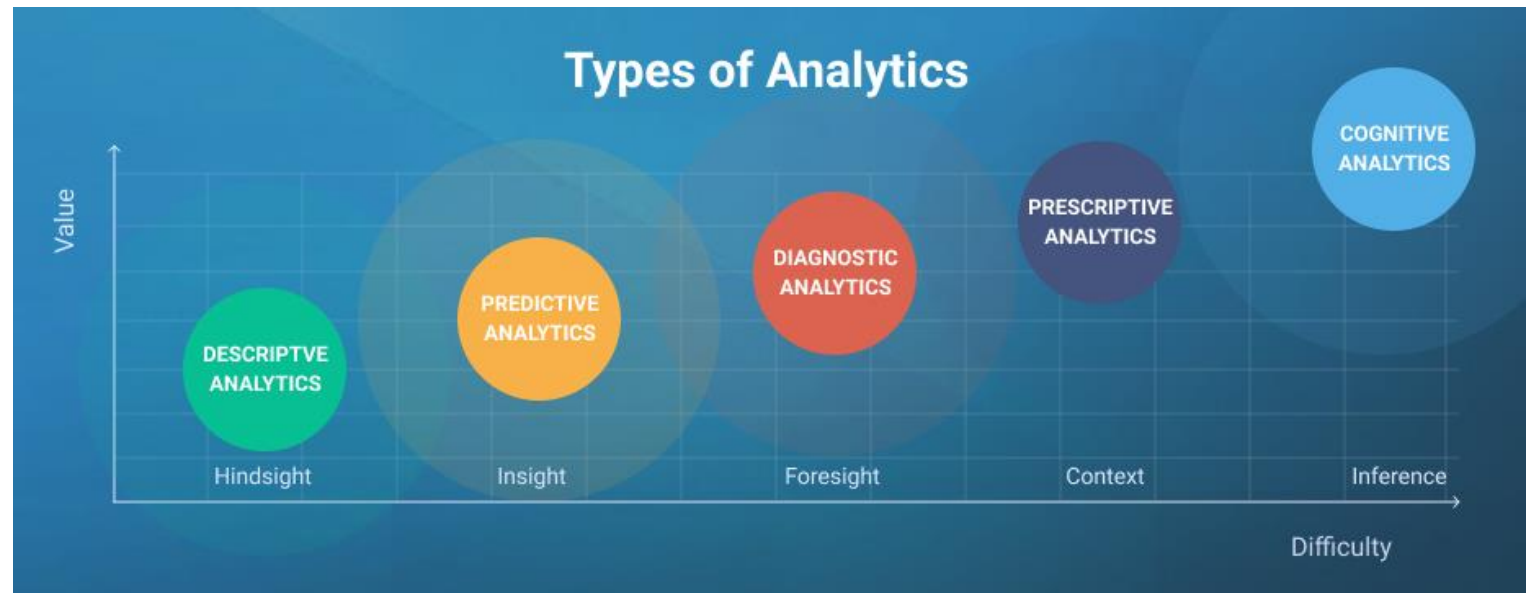
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These activities are categorized as:

- descriptive,
- diagnostic,
- predictive,
- prescriptive, and
- cognitive analytics.



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- Descriptive analytics answers questions about **what happened**, based on historical data, to inform decisions about the future.

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Descriptive

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Diagnostic

- Diagnostic analytics helps answer questions about **why things happened**.

Understand data analytics types



Predictive

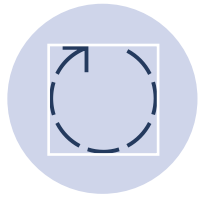
- Predictive analytics helps answer questions about **what will happen** in the future.

Understand data analytics types



Predictive

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Prescriptive

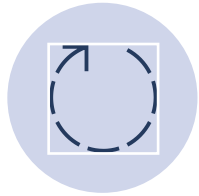
- Prescriptive analytics takes predictive analytics one step further and helps answer questions about what **actions should be taken** to achieve a goal or target.

Understand data analytics types



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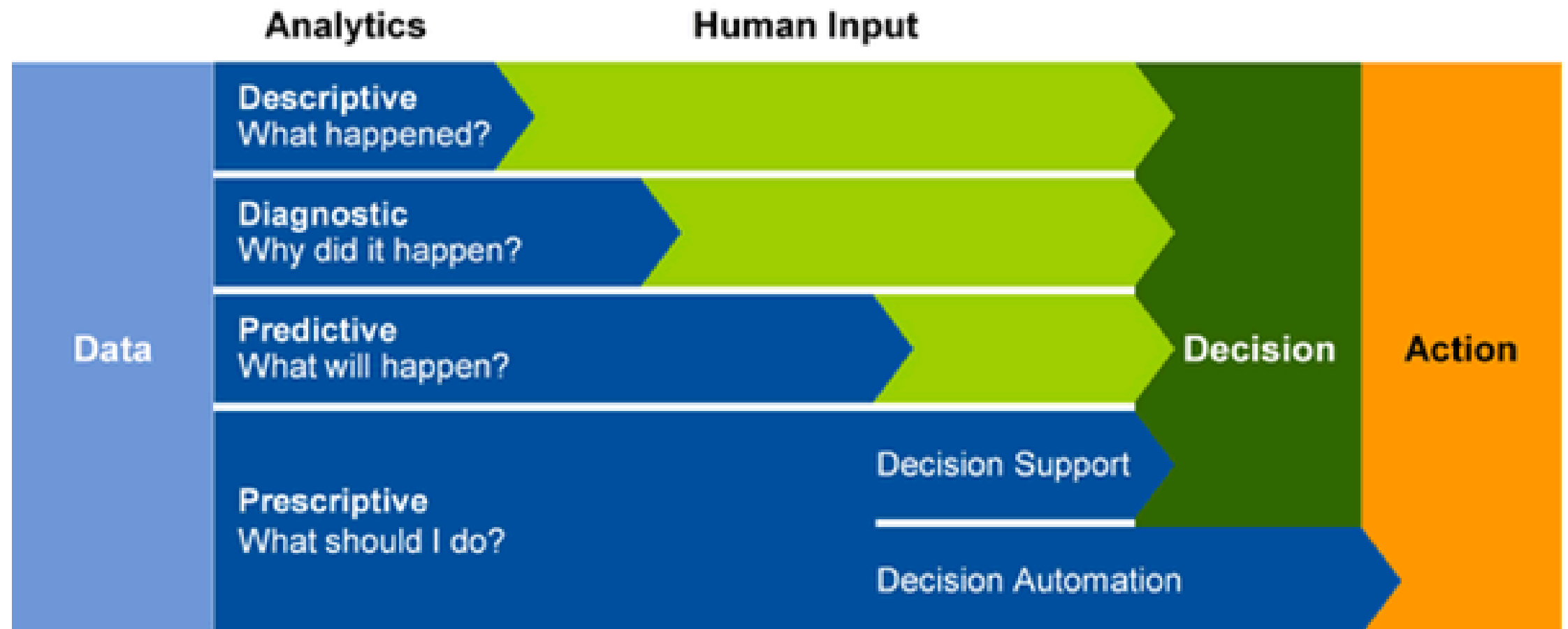
Prescriptive

- Prescriptive analytics takes predictive analytics one step further and helps answer questions about what **actions should be taken** to achieve a goal or target.



Cognitive

- Cognitive analytics **combines artificial intelligence, machine learning, and data analytics approaches** to guide decision making.



Explore the data analytics process

Data analytics is the process of **collecting**, **transforming**, and **presenting** data to inform decision making.

Explore the data analytics process

Data analytics is the process of **collecting**, **transforming**, and **presenting** data to inform decision making.

The **analytics process is cyclical by nature**.

Exposing data and insights often leads to requests for **more analysis**, which leads to **more feedback**, and so on.



Explore the data analytics process

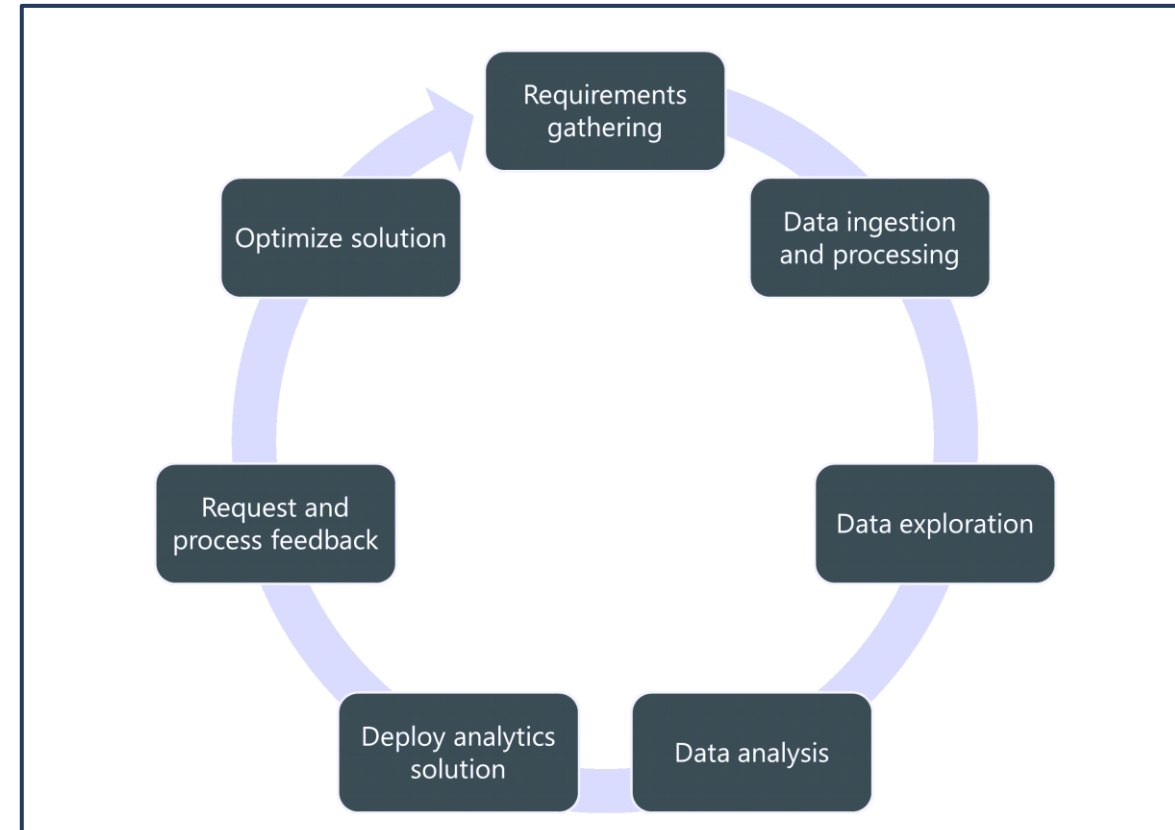
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Steps of data analysis process:

- Requirements gathering
- Data ingestion and processing
- Data exploration
- Data analysis
- Deploy analytics solution
- Request and process feedback
- Optimize solution
- Begin again



Understanding how data is structured and stored is a **critical step** that occurs at the beginning of every analytics project, during requirements gathering.

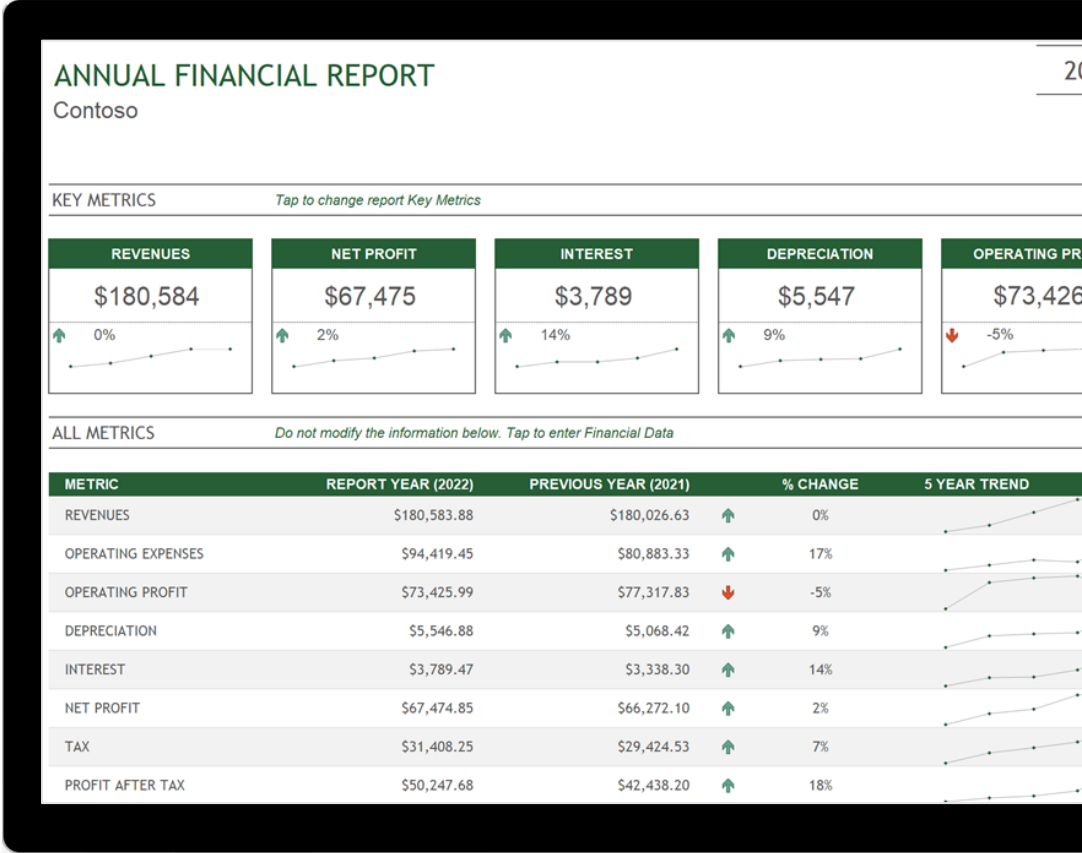
Both **structured and unstructured** data are suitable for analysis, but the **tools** the data team will use to ingest, transform, and store data **will differ** according to the data type.



Understand types of data and data storage (1/2)

Structured data:

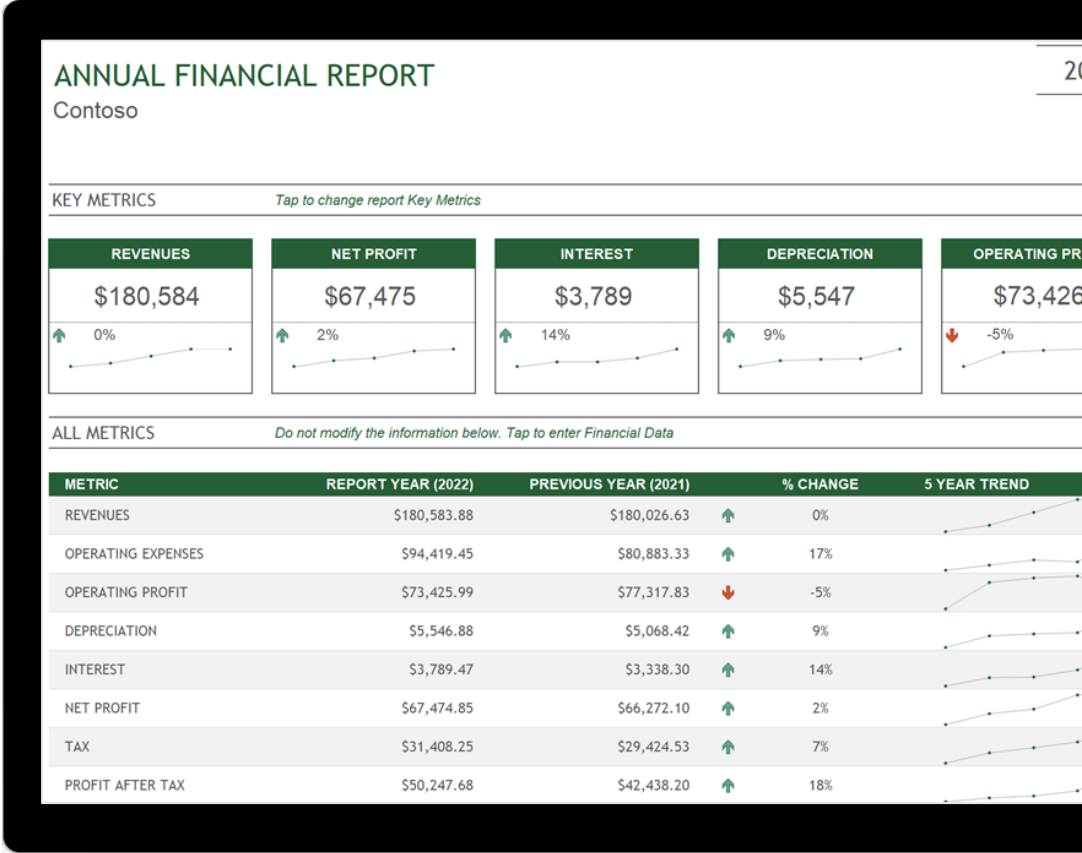
- Letters and numbers organized into columns and rows for simplified search and processing.



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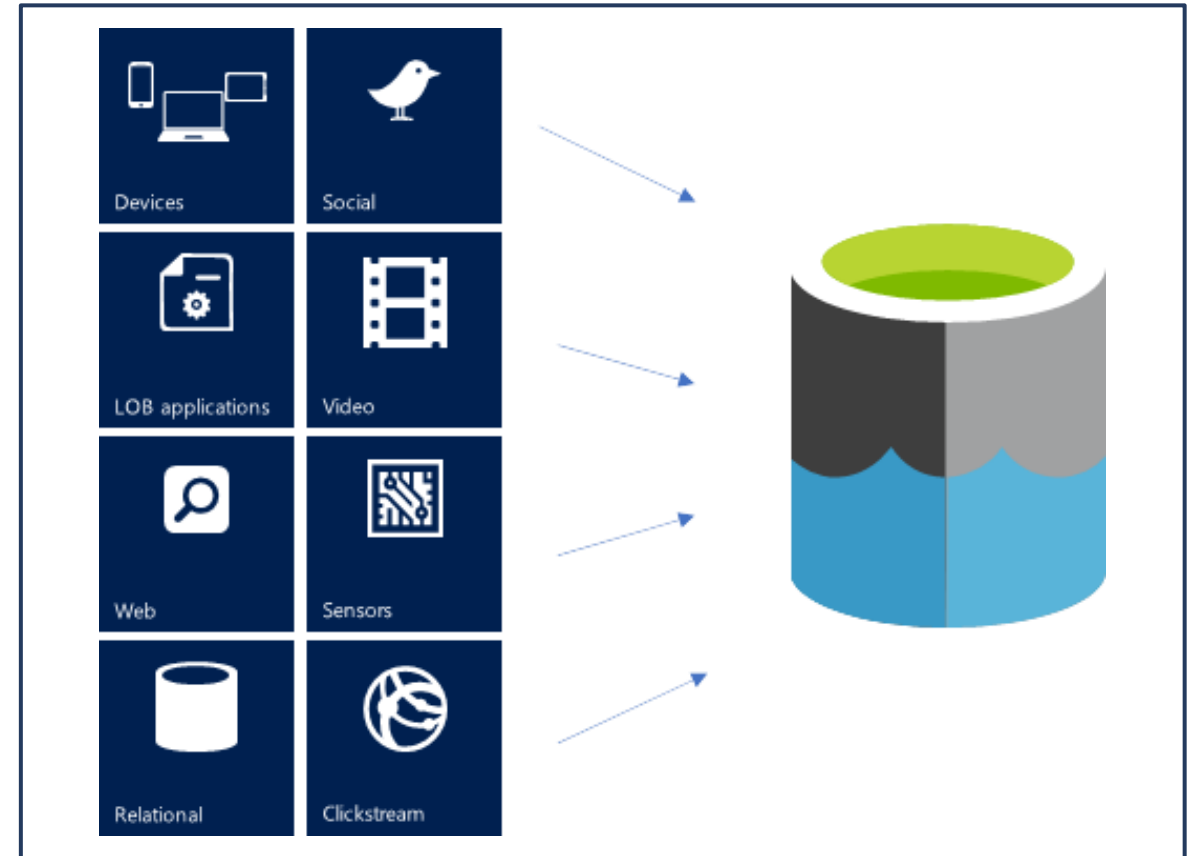
- Letters and numbers organized into columns and rows for simplified search and processing.
- Lends well to **all types of analytics** and is the most accessible.



Understand types of data and data storage (2/2)

Unstructured data:

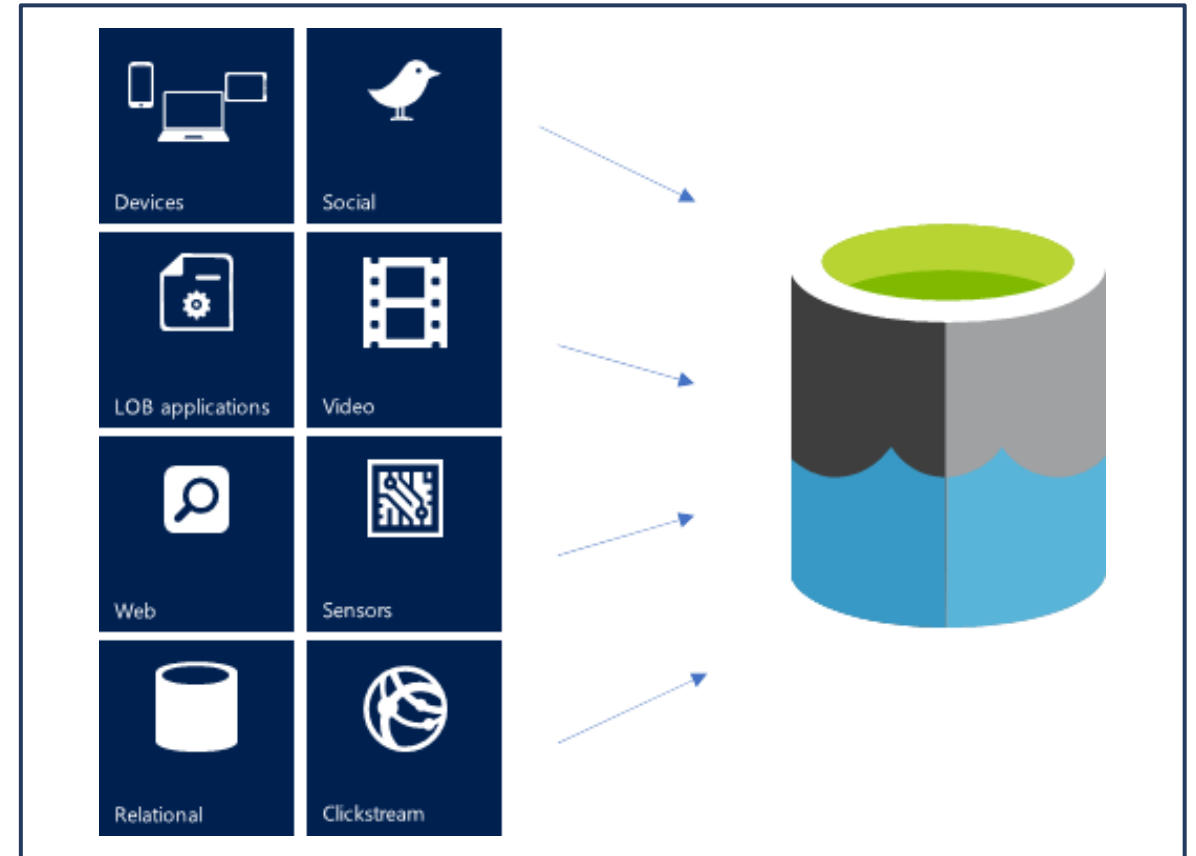
- Information that isn't organized in any discernable manner.



Understand types of data and data storage (2/2)

Unstructured data:

- Information that isn't organized in any discernable manner.
- Often more **suitable for qualitative analysis** and is stored in **non-relational databases** and data lakes.



Knowledge check (1/2)



Which of the following is an example of descriptive analytics?



- A monthly sales report looking at sales data over the last year.
- A social media algorithm that recommends curated content.
- An annual HR report that forecasts predicted attrition for the next year.



What are the first three steps in the data analytics process?

- Data exploration, data analysis, and deploy analytics solution.
- Data analysis, deploy analytics solution, and request feedback.



Requirements gathering, data ingestion and processing, and data exploration.

Knowledge check (2/2)



Why is it important to understand the difference between structured and unstructured data?

- There is no difference between structured and unstructured data.
-  Understanding the difference can determine where data should be stored and what kind of analysis is most appropriate.
- Understanding the difference will determine if the business requirements have been met.

Summary

In this section, you've learned how to:

- Define the five types of data analytics
- Describe the data analytics process
- Identify data types and storage

To learn more:

- [Explore Azure database and analytics services](#)
- [Explore fundamentals of modern data warehousing](#)

Explore data analytics at scale



Introduction

In this section, you will:

- Explore **job roles** in analytics
- Understand **tools for scaling** analytics solutions



[REMEMBER]

- **Large** data projects can be complex.
- The projects often involve **hundreds of decisions**.
- **Multiple people** are typically involved, and each person helps take the project from design to production.
- Your organization **may define roles differently** or give them different names.

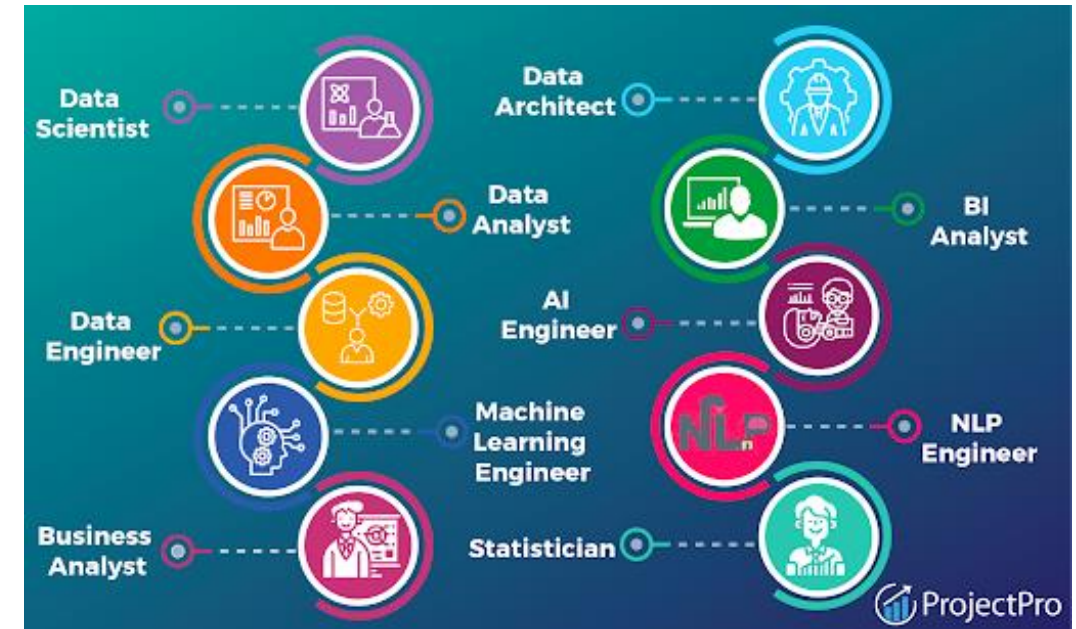


Data team roles and responsibilities

The roles described in this unit represent the most common **division of tasks and responsibilities**.

These roles are common **in modern data projects**:

- Data analyst
- Database administrator
- Data engineer
- Data scientist
- Data architect



Data team roles and responsibilities



Data Analyst

Enables businesses to maximize the value of their data assets.

Responsible for

- exploring data to identify trends,
- designing and building analytical models, and
- enabling advanced analytics capabilities through reports and visualizations.

A data analyst **processes raw data into relevant insights based on identified business requirements.**

Data team roles and responsibilities



Database Administrator

Responsible for:

- the design,
- implementation,
- maintenance, and
- operational aspects of

on-premises and cloud-based database systems.

Data team roles and responsibilities



Data Engineer

- Typically configure data platform technologies that are on-premises and in the cloud.
- Manage and secure the flow of structured and unstructured data from multiple sources.

Data team roles and responsibilities



- Perform advanced analytics to extract value from data.
- Work can vary from descriptive analytics to predictive analytics.

Data Scientist

Data team roles and responsibilities



[Typically] Responsible for:

- planning and executing an overall data management strategy, including defining standards of data quality and security.

Data Architect

Tasks and tools for data analysts



??

Data analyst tasks:

- Collect and clean data for analysis
- Ingest, transform, and model data in a reporting tool



Synapse



Power BI



Purview

Data analyst Tools

Tasks and tools for data analysts

Data analyst tasks:

- Collect and clean data for analysis
- Ingest, transform, and model data in a reporting tool
- Generate data products, dashboards and reports
- Communicate findings with relevant stakeholders



Synapse



Power BI



Purview

Data analyst Tools

Tasks and tools for data analysts

Data analyst tasks:

- Collect and clean data for analysis
- Ingest, transform, and model data in a reporting tool
- Generate data products, dashboards and reports
- Communicate findings with relevant stakeholders
- Monitor usage of analytics solution
- Gather and incorporate feedback and additional requirements



Synapse



Power BI



Purview

Data analyst Tools

Scale analytics with Azure Synapse Analytics and Power BI

Recent shift in terminology from **business intelligence to modern analytics** can be attributed to the sheer **volume of data generated** that needs to be **processed quickly**. This wouldn't be possible with a **traditional ETL**, data warehousing, and reporting approach.



Scale analytics with Azure Synapse Analytics and Power BI

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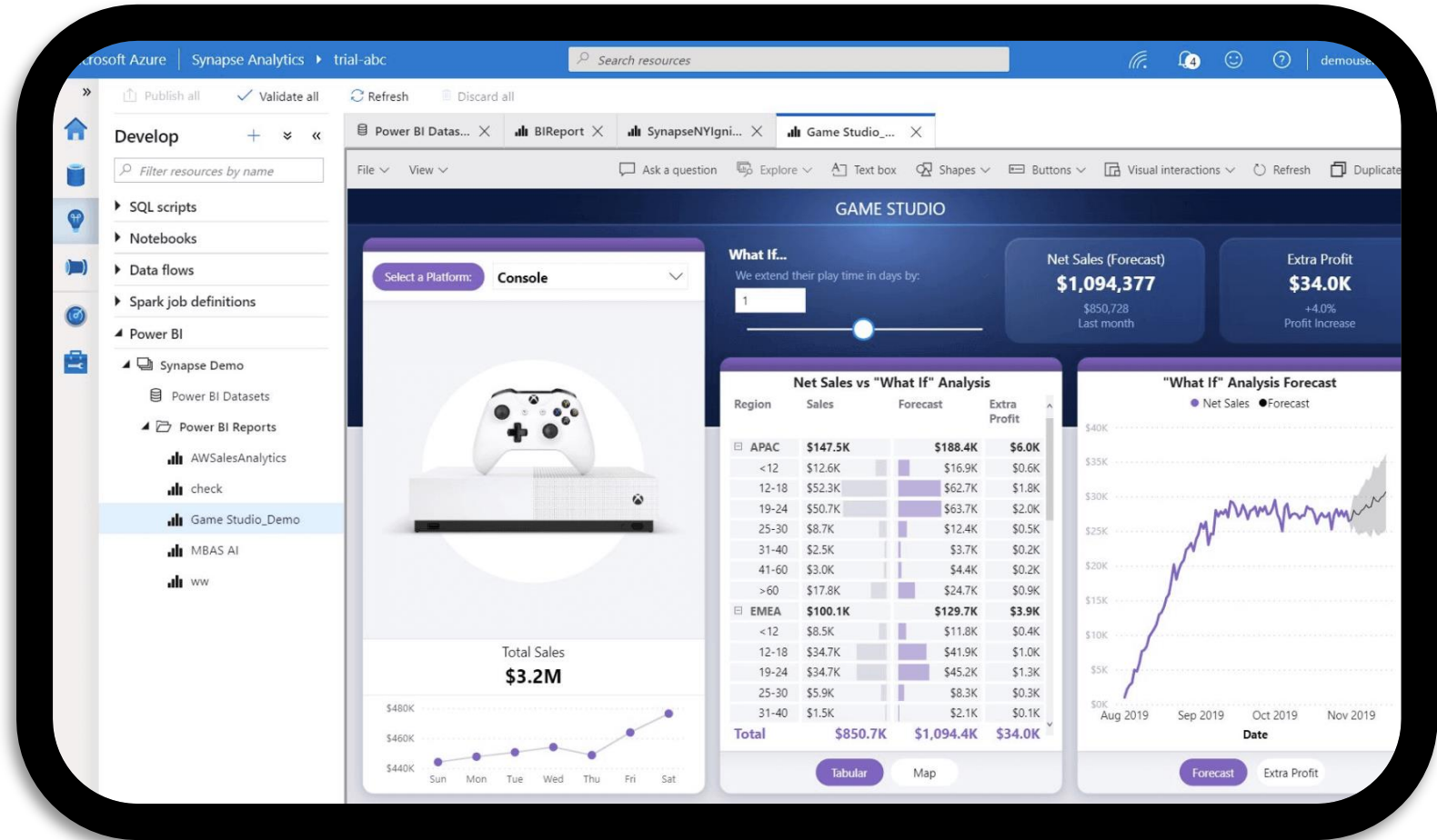
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[From the previous lectures]

- We know that with the power of Azure Synapse Analytics we are able to connect to and process data stored in large scale Data Warehousing databases, query data in Azure Storage, Azure Data Lake Gen1 and 2, and Cosmos DB using Serverless SQL Pools with familiar SQL commands.

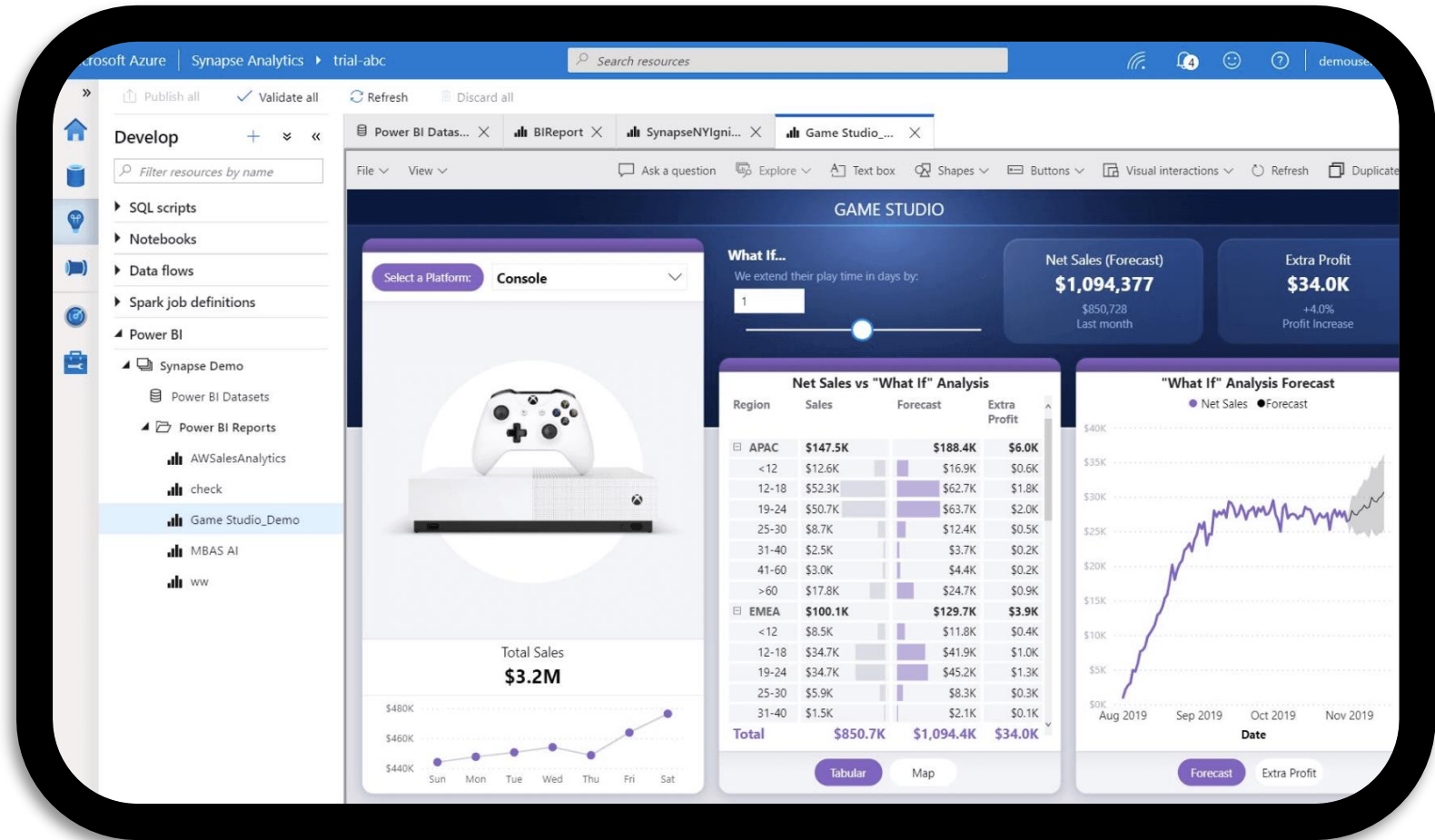
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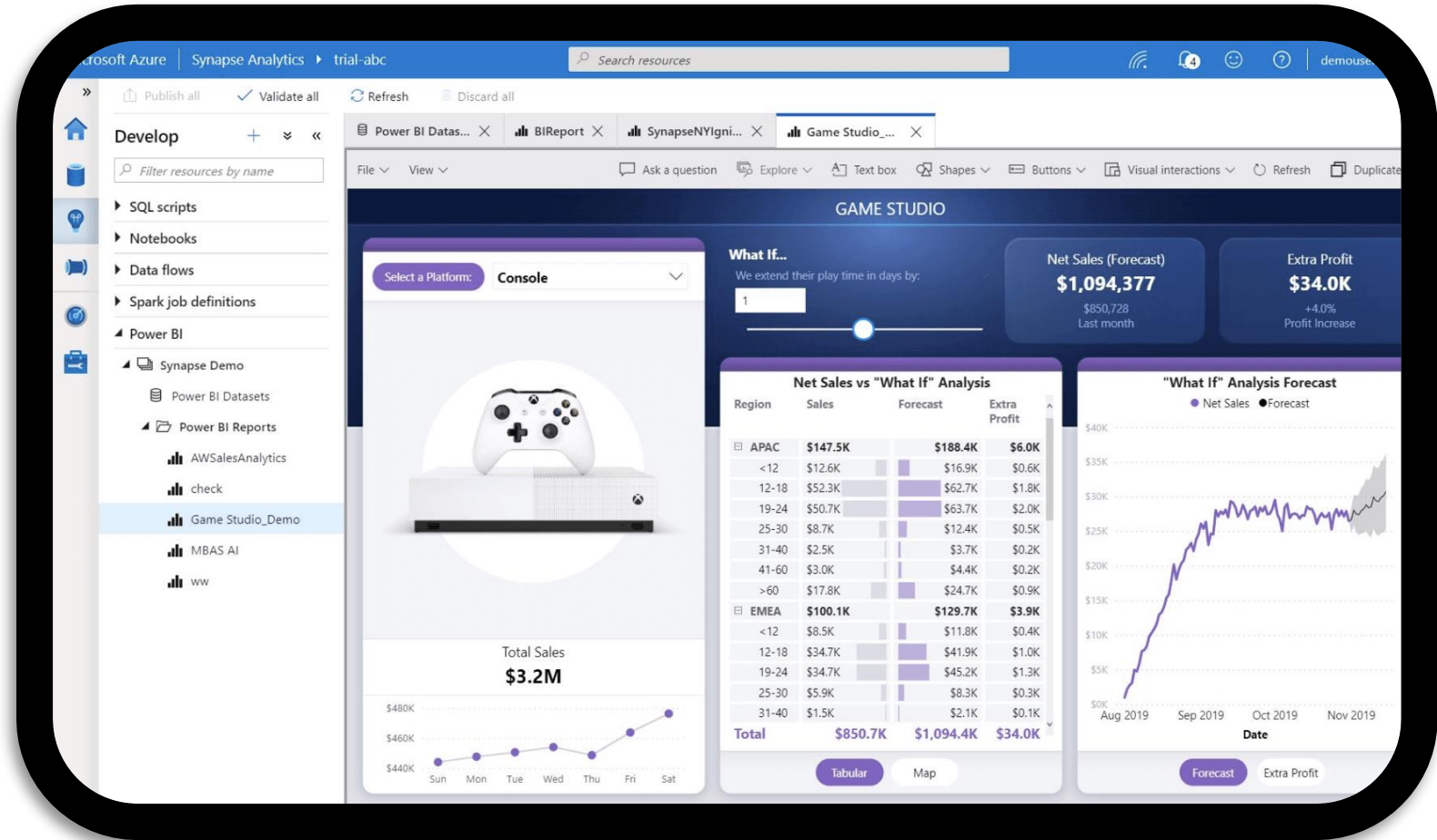
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- Using **Power BI** with Synapse enables analysts to process large-scale data quickly.
- **Power BI and Synapse are natively integrated.**



Strategies to scale analytics

Scaling analytics and data use across an enterprise *isn't possible with just great tools.*

This requires a **cultural shift** in which the **technology** becomes the **easy piece** to implement.

Strategies to scale analytics

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THE **DIFFICULT** PART IS!

- **Building a data culture,**
- **creating a community of practice** and
- **implementing a data governance strategy**

These are tangible ways to focus on people and process.

Strategies to scale analytics

Several tangible ways to focus on people and process:

Build a data culture



- Building and maintaining a data culture is a key aspect in an organization's digital transformation.
- It encourages a culture that makes data-informed decisions.
- It's important that you have a well-understood definition of what a healthy data culture means to your organization.

Strategies to scale analytics

Several tangible ways to focus on people and process:

Establish a Center of Excellence (CoE)



An analytics *Center of Excellence (COE)* is:

- an internal team in your organization whose role is to maximize the potential of analytics in the organization.
- traditionally made up of both technical and business experts who evangelize the data-driven culture and guide others in the uptake of analytics solutions.
- provide expert advice to others and serve as a guiding force promoting the uptake of analytics solutions.

Strategies to scale analytics

Several tangible ways to focus on people and process:

Build a community of practice



- A *community of practice* is a group of people with a common interest that interacts with, and helps, each other on a voluntary basis.
- Mastering analytics tools like Power BI creates common interests and encourages individuals across departments and teams to work towards a common goal.
- Building a community of practice at your organization encourages knowledge sharing and can catalyze the building of a healthy data culture from the ground-up.

Knowledge check (1/2)



Which of the following tasks are completed by a data analyst?



Build monthly sales reports and gather feedback from the business on how reports are being used.

- Design, implement, and maintain operational aspects of on-premises and cloud-based database systems.
- Manage and secure the flow of structured and unstructured data from multiple sources into analytical data storage.



What Microsoft tools do data analysts use for enterprise-scale analytics solutions?

- Azure Machine Learning and Azure Cosmos DB.
- Microsoft Excel.



Azure Synapse Analytics and Microsoft Power BI.

Knowledge check (2/2)



What people-focused strategies can help an organization achieve analytics at scale?

- Azure Synapse Analytics and Microsoft Power BI can help achieve analytics at scale in an organization.
- ➔ Building a data culture, establishing a Center of Excellence, and creating a community of practice.
- Achieving analytics at scale should be technology focused rather than people focused.

Summary

In this section you've learned how to:

- Describe job roles in analytics
- Understand tools for scaling analytics solutions

To learn more:

- [Get started with Azure Synapse Analytics](#)
- [Power BI adoption roadmap](#)

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Thank You!